

Assessment 2 (Project)

Multi-Site Network Design and Implementation

Industry	Education Service
Network Block	172.31.0.0/19 (8,192 addresses)
Locations	8 (7 branch sites + 1 Data Centre)
Tools	Cisco Packet Tracer

Name – Mohammed Jonaed Haque

Student ID – 48600776

Unit Code – COMP2270

1. Client Requirement Summary

The client is an Education Services organization operating across seven geographically distributed branch sites and a centralized Data Centre. All design decisions – subnetting, topology, IP assignments, and routing - are based exclusively on the personalized dataset assigned from iLearn database.

1.1 Assigned Database Values

Parameter	Assigned Value
Industry	Education Services
Network Address Block	172.31.0.0/19 (8,192 total addresses 8,190 usable hosts 172.31.0.0 – 172.31.31.255)
Number of Locations	8 (7 branch sites + 1 Data Centre)
Data Centre	2 servers – Server1 and Server2

1.2 Users Per Location

Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Data Centre
21	24	28	14	29	23	80	2 Servers

1.3 Infrastructure Requirements

- Each site has a dedicated Cisco 2960-24TT site switch connecting all local PCs.
- Sites 1-4 aggregate through Switch1 to Router1 (ISR4321) via a /29 transit subnet.
- Sites 5-7 aggregate through Switch0 to Router2 (Cisco 2911) via a /29 transit subnet.
- MainRouter (Cisco 2911) forms a redundant triangle core with Router1 and Router2.
- Data Centre is connected via Router2 on a dedicated /29 subnet – Server1 and Server2 on Switch2.
- Three point-to-point /30 links (two GigabitEthernet, one Serial) interconnect the core routers.
- A dedicated site router (Cisco 2911) was deployed at each of the 7 sites – justified in Section 1.4.

1.4 Approved Design Variation – Site Routers

The original specification lists 3 routers. However, Cisco Packet Tracer does not support the IP address **secondary** command in the installed version, making it impossible to assign a correct VLSM gateway IP to each individual site subnet when multiple sites share one aggregation uplink interface. **A dedicated Cisco 2911 site router was therefore deployed between each site switch and its aggregation switch, providing the precise gateway IP for each subnet.**

This approach was explicitly approved by the Teaching Assistant on the iLearn Discussion Forum. When asked whether additional routers could be used (3 main routers + 7 site routers). Jayden King replied: “Yes, you can use additional routers.” (Jayden King, 15 May 2026).

Assesment 2 - secondary IP command

Display replies in nested form
Settings


Assesment 2 - secondary IP command
 by Mohammed Jonaed Haque - Thursday, 14 May 2026, 10:53 PM

The specification requires separate subnets for Sites 1–4 through one Router1 interface, but Packet Tracer is rejecting the secondary IP command, and VLAN/trunking is stated as out of scope. Should we use secondary IPs, extra router interfaces, or a simplified subnet for the aggregation group?

[Permalink](#)
[Reply](#)


Re: Assesment 2 - secondary IP command
 by Mohammed Jonaed Haque - Friday, 15 May 2026, 12:53 AM

or am i allowed to use additional routers for each sites. (3main routers + 7 site routers), this might help to solve the problem

[Permalink](#)
[Show parent](#)
[Reply](#)


Re: Assesment 2 - secondary IP command
 by Jayden King - Friday, 15 May 2026, 1:41 PM

Yes, you can use additional routers.

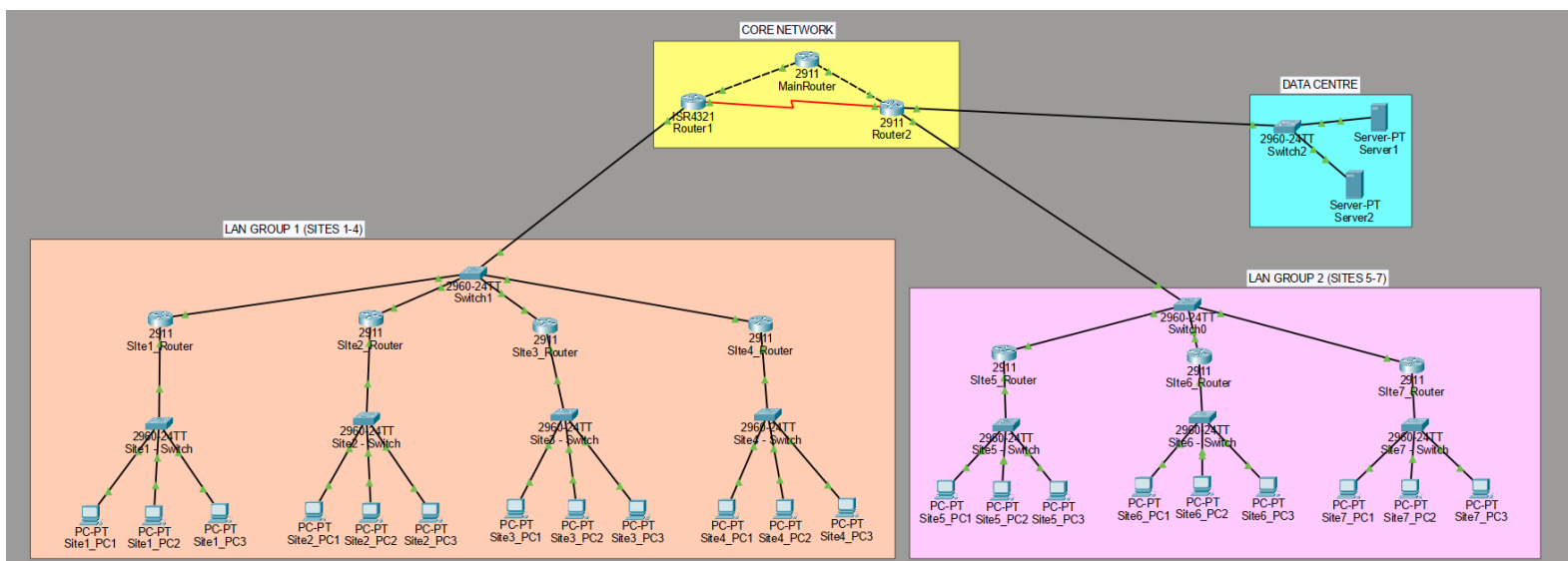
[Permalink](#)
[Show parent](#)
[Reply](#)

Each site router uses a single default route pointing to the aggregation router. Keeping the routing design simple and consistent with the static routing scope of this project. The specification clause – “Alternative valid designs or topologies are also acceptable, provided they meet the assessment requirements and demonstrate correct configuration and connectivity” – explicitly permits this design.

2. Network Topology

2.1 Full Topology Screenshot

The complete logical topology from Cisco Packet Tracer is shown below, organized into 4 clearly labeled and color-coded zones as required by the specification.



2.2 Zone Summary

Zone	Devices	Connectivity
Core Network	MainRouter, Router1, Router2	Triangle: 2 x GigabitEthernet /30 + 1 x Serial /30
LAN Group 1 (Sites 1– 4)	Switch1, Site1_Router – Site4_Router, Site1_Switch – Site4_Switch, PCs 1-12	Site routers connect via /29 transit subnet through Switch1 to Router1
LAN Group 2 (Sites 5– 7)	Switch0, Site5_Router – Site7_Router, Site5_Switch – Site7_Switch, PCs 13-21	Site routers connect via /29 transit subnet through Switch0 to Router2
Data Centre	Switch2, Server1, Server2	Router2 GigabitEthernet0/1 → Switch2 → Server1, Server2

2.3 Complete Device Inventory

Hostname	Model	Role
MainRouter	Cisco 2911	Core triangle router – GigabitEthernet links to Router1 (Gi0/0) and Router2 (Gi0/1)
Router1	Cisco ISR4321	Core triangle router - aggregates Sites 1-4 via transit subnet on Gi0/0/1
Router2	Cisco 2911	Core triangle router - aggregates Sites 5-7 (Gi0/2) and Data Centre (Gi0/1)
Site1_Router – Site4_Router	Cisco 2911 x4	Site routers: provide per- subnet default gateway; transit interface to Switch1
Site5_Router – Site7_Router	Cisco 2911 x3	Site routers: provide per- subnet default gateway; transit interface to Switch0
Switch1	Cisco 2960-24TT	Aggregation switch – Sites 1-4 transit subnet 172.31.1.152/29
Switch0	Cisco 2960-24TT	Aggregation switch – Sites 5-7 transit subnet 172.31.1.160/29
Switch2	Cisco 2960-24TT	Data Centre access switch – subnet 172.31.1.128/29
Site1_Switch – Site7_Switch	Cisco 2960-24TT x 7	Site access switches – connect PCs to site router LAN interface
Site1_PC1 – Site7_PC3	PC_PT x 21	3 representative user PCs per site (minimum 3 as per specification)
Server1, Server2	Server-PT x2	Centralized Data Centre servers providing shared services.

3. VLSM Subnetting – Working and Address Plan

3.1 Why VLSM was used

Variable Length Subnet Masking was selected over equal-size subnets because user **counts vary dramatically across sites** from 14 users at Site 4 to 80 users at Site 7. Allocating a uniform /25 (the size needed for Site 7) to all locations would consume $7 \times 128 = 896$ addresses for site LANs alone, creating severe waste and risking exhaustion of the /19 block once transit and P2P subnets are included. **VLSM solves this by allocating the minimum sufficient subnet at each location**, resulting in a total consumption of fewer than 600 addresses - leaving substantial reserve within the 8,192-address /19 block.

3.2 Subnet Sizing Methodology

Each site subnet must accommodate three mandatory addresses categories as required by the specification:

Required usable hosts = Users + 1 (default gateway) + 1 (switch management IP)

The switch management IP must be a valid IP from the site subnet (spec Section 6c). This raises the minimum host requirement above the raw user count, directly influencing subnet size decisions for **Sites 5 and 3**, as shown in the table below.

3.3 Subnet Size Selection

Site	User	+Gateway	+Switch Mgmt	= Total	/27 30 hosts	/26 62 hosts	/25 126 hosts	Decision	Pfx
Site 7	80	+1	+1	82	X Too Small	X Too small	✓ Fits	Minimum Fit	/25
Site 5	29	+1	+1	31	X $30 < 31$	✓ Fits	—	Required	/26 ★
Site 3	28	+1	+1	30	✓ Exact fit	✓ Chosen for safety	—	Zero spare	/26 ★★
Site 2	24	+1	+1	26	✓	—	—	Minimum Fit	/27
Site 6	23	+1	+1	25	✓ Fits	—	—	Minimum Fit	/27

Site 1	21	+1	+1	23	✓ Fits	—	—	Minimum Fit	/27
Site 4	14	+1	+1	16	✓ Fits	—	—	Minimum Fit	/27
DC	2	+1	+1	4	—	—	—	Use /29 (6 hosts)	/29

★ Site 5 (29 users): $29+1+1 = 31$ minimum hosts. A /27 provides only 30 usable addresses mathematically insufficient. /26 (62 usable) is the correct minimum allocation.

★★ Site 3 (28 users): $28+1+1 = 30$ minimum hosts, exactly exhausting a /27 with zero spare capacity. /26 was chosen to prevent subnet exhaustion from any additional management or monitoring device.

3.4 Complete VLSM Address Plan → Ordered Largest-First

All 13 subnets allocated from 172.31.0.0/19, presented in strict largest-to-smallest order. Transit subnet and the Data Centre /29 are placed before P2P /30 links as required by VLSM best practice.

#	Location	Pfx	Network	Subnet Mask	First Host	Last Host	Default Gateway	Hosts	Broadcast
1	Site 7	/25	172.31.0.0	255.255.255.128	172.31.0.1	172.31.0.126	172.31.0.1	126	172.31.0.127
2	Site 5	/26	172.31.0.128	255.255.255.192	172.31.0.129	172.31.0.190	172.31.0.129	62	172.31.0.191
3	Site 3	/26	172.31.0.192	255.255.255.192	172.31.0.193	172.31.0.254	172.31.0.193	62	172.31.0.255
4	Site 2	/27	172.31.1.0	255.255.255.224	172.31.1.1	172.31.1.30	172.31.1.1	30	172.31.1.31
5	Site 6	/27	172.31.1.32	255.255.255.224	172.31.1.33	172.31.1.62	172.31.1.33	30	172.31.1.63
6	Site 1	/27	172.31.1.64	255.255.255.224	172.31.1.65	172.31.1.94	172.31.1.65	30	172.31.1.95
7	Site 4	/27	172.31.1.96	255.255.255.224	172.31.1.97	172.31.1.126	172.31.1.97	30	172.31.1.127
8	Data Centre	/29	172.31.1.128	255.255.255.248	172.31.1.129	172.31.1.134	172.31.1.129	6	172.31.1.135
9	Transit SW1	/29	172.31.1.152	255.255.255.248	172.31.1.153	172.31.1.158	172.31.1.153 (R1)	6	172.31.1.159
10	Transit SW0	/29	172.31.1.160	255.255.255.248	172.31.1.161	172.31.1.166	172.31.1.161	6	172.31.1.167
11	R0 ↔ R1 Link	/30	172.31.1.136	255.255.255.252	172.31.1.137	172.31.1.138	—	2	172.31.1.139

1 2	R0 ↔ R2 Link	/3 0	172.31.1. 140	255.255.255. 252	172.31.1. 141	172.31.1. 142	—	2	172.31.1. 143
1 3	R1 ↔ R2 Link	/3 0	172.31.1. 144	255.255.255. 252	172.31.1. 145	172.31.1. 146	—	2	172.31.1. 147

4. Configuration Summary

All CLI configurations were entered directly in Cisco Packet Tracer. Key commands are presented below for each device category. Full interface details are in Appendix A.

4.1 MainRouter (R0) – Cisco 2911

Two native GigabitEthernet interfaces form P2P links to Router1 and Router2. Static routes direct all 13 subnets to the correct next-hop router.

```
hostname MainRouter
```

```
!
```

```
interface GigabitEthernet0/0
```

```
description P2P link to Router1
```

```
ip address 172.31.1.137 255.255.255.252
```

```
no shutdown
```

```
!
```

```
interface GigabitEthernet0/1
```

```
description P2P link to Router2
```

```
ip address 172.31.1.141 255.255.255.252
```

```
no shutdown
```

```
!
```

```
!-- Static routes: Sites 1-4 and Transit SW1 via Router1 (172.31.1.138)
```

```
ip route 172.31.1.64 255.255.255.224 172.31.1.138
```

```
ip route 172.31.1.0 255.255.255.224 172.31.1.138
```

```
ip route 172.31.0.192 255.255.255.192 172.31.1.138
```

```
ip route 172.31.1.96 255.255.255.224 172.31.1.138
```

```
ip route 172.31.1.152 255.255.255.248 172.31.1.138
```

```
!-- Sites 5-7, Transit SW0, and Data Centre via Router2 (172.31.1.142)
```

```
ip route 172.31.0.0 255.255.255.128 172.31.1.142
ip route 172.31.0.128 255.255.255.192 172.31.1.142
ip route 172.31.1.32 255.255.255.224 172.31.1.142
ip route 172.31.1.128 255.255.255.248 172.31.1.142
ip route 172.31.1.160 255.255.255.248 172.31.1.142
ip route 172.31.1.144 255.255.255.252 172.31.1.138
end
write memory
```

4.2 Router1(R1) – Cisco ISR4321

Three Interfaces: Gi0/0/0 to MainRouter, Gi0/0/1 as the Transit SW1 gateway, Serial0/1/0 to Router2. Routes point to individual site routers for Sites 1-4.

```
hostname Router1
!
interface GigabitEthernet0/0/0
description P2P link to MainRouter
ip address 172.31.1.138 255.255.255.252
no shutdown
!
interface GigabitEthernet0/0/1
description Transit SW1 – aggregation for Sites 1-4
ip address 172.31.1.153 255.255.255.248
no shutdown
!
interface Serial0/1/0
```

```
description Serial P2P to Router2
ip address 172.31.1.145 255.255.255.252
no shutdown
!
! -- Sites 1-4: route to each site router individually
ip route 172.31.1.64 255.255.255.224 172.31.1.154
ip route 172.31.1.0 255.255.255.224 172.31.1.155
ip route 172.31.0.192 255.255.255.192 172.31.1.156
ip route 172.31.1.96 255.255.255.224 172.31.1.157
! -- Remote subnets via Router2 serial (172.31.1.146)
ip route 172.31.0.0 255.255.255.128 172.31.1.146
ip route 172.31.0.128 255.255.255.192 172.31.1.146
ip route 172.31.1.32 255.255.255.224 172.31.1.146
ip route 172.31.1.128 255.255.255.248 172.31.1.146
ip route 172.31.1.160 255.255.255.248 172.31.1.146
ip route 172.31.1.140 255.255.255.252 172.31.1.137
end
write memory
```

4.3 Router2 (R2) – Cisco 2911

Four interfaces: Gi0/0 to MainRouter, Gi0/1 to Data Centre, Gi0/2 as Transit SW0 gateway, Serial to Router1.

```
hostname Router2
!
interface GigabitEthernet0/0
description P2P link to MainRouter
ip address 172.31.1.142 255.255.255.252
no shutdown
!
interface GigabitEthernet0/1
description Data Centre gateway
ip address 172.31.1.129 255.255.255.248
no shutdown
!
interface GigabitEthernet0/2
description Transit SW0 – aggregation for Sites 5-7
ip address 172.31.1.161 255.255.255.248
no shutdown
!
interface Serial0/0/0
description Serial P2P to Router1
ip address 172.31.1.146 255.255.255.252
no shutdown
!
! -- Sites 5-7: route to each site router individually
ip route 172.31.0.128 255.255.255.192 172.31.1.162
```

```
ip route 172.31.1.32 255.255.255.224 172.31.1.163
ip route 172.31.0.0 255.255.255.128 172.31.1.164
!-- Remote subnets via Router1 serial (172.31.1.145)
ip route 172.31.1.64 255.255.255.224 172.31.1.145
ip route 172.31.1.0 255.255.255.224 172.31.1.145
ip route 172.31.0.192 255.255.255.192 172.31.1.145
ip route 172.31.1.96 255.255.255.224 172.31.1.145
ip route 172.31.1.152 255.255.255.248 172.31.1.145
ip route 172.31.1.136 255.255.255.252 172.31.1.141
end
write memory
```

4.4 Site Routers - Full Configuration (All 7)

All 7 site routers follow an identical pattern: one LAN interface with the site VLSM gateway IP, one transit interface to the aggregation switch, and a single default route upstream. The complete verified CLI configuration for each router is shown below as confirmed by show running-config in Packet Tracer.

Site1_Router — Site 1 subnet 172.31.1.64/27

```
hostname Site1_Router
interface GigabitEthernet0/0
ip address 172.31.1.65 255.255.255.224
duplex auto
speed auto
no shutdown
interface GigabitEthernet0/1
```

```
ip address 172.31.1.154 255.255.255.248
```

```
duplex auto
```

```
speed auto
```

```
no shutdown
```

```
ip classless
```

```
ip route 0.0.0.0 0.0.0.0 172.31.1.153
```

```
end
```

```
write memory
```

Site2_Router — Site 2 subnet 172.31.1.0/27

```
hostname Site2_Router
```

```
interface GigabitEthernet0/0
```

```
ip address 172.31.1.1 255.255.255.224
```

```
duplex auto
```

```
speed auto
```

```
no shutdown
```

```
interface GigabitEthernet0/1
```

```
ip address 172.31.1.155 255.255.255.248
```

```
duplex auto
```

```
speed auto
```

```
no shutdown
```

```
ip classless
```

```
ip route 0.0.0.0 0.0.0.0 172.31.1.153
```

```
end
```

```
write memory
```

Site3_Router — Site 3 subnet 172.31.0.192/26

```
hostname Site3_Router
```

```
interface GigabitEthernet0/0
ip address 172.31.0.193 255.255.255.192
duplex auto
speed auto
no shutdown
interface GigabitEthernet0/1
ip address 172.31.1.156 255.255.255.248
duplex auto
speed auto
no shutdown
ip classless
ip route 0.0.0.0 0.0.0.0 172.31.1.153
end
write memory
```

Site4_Router — Site 4 subnet 172.31.1.96/27

```
hostname Site4_Router
interface GigabitEthernet0/0
ip address 172.31.1.97 255.255.255.224
duplex auto
speed auto
no shutdown
interface GigabitEthernet0/1
ip address 172.31.1.157 255.255.255.248
duplex auto
speed auto
no shutdown
ip classless
```

```
ip route 0.0.0.0 0.0.0.0 172.31.1.153
```

```
end
```

```
write memory
```

Site5_Router — Site 5 subnet 172.31.0.128/26

```
hostname Site5_Router
```

```
interface GigabitEthernet0/0
```

```
ip address 172.31.0.129 255.255.255.192
```

```
duplex auto
```

```
speed auto
```

```
no shutdown
```

```
interface GigabitEthernet0/1
```

```
ip address 172.31.1.162 255.255.255.248
```

```
duplex auto
```

```
speed auto
```

```
no shutdown
```

```
ip classless
```

```
ip route 0.0.0.0 0.0.0.0 172.31.1.161
```

```
end
```

```
write memory
```

Site6_Router — Site 6 subnet 172.31.1.32/27

```
hostname Site6_Router
```

```
interface GigabitEthernet0/0
```

```
ip address 172.31.1.33 255.255.255.224
```

```
duplex auto
```

```
speed auto
```

```
no shutdown
```



```
interface GigabitEthernet0/1
ip address 172.31.1.163 255.255.255.248
duplex auto
speed auto
no shutdown
ip classless
ip route 0.0.0.0 0.0.0.0 172.31.1.161
end
write memory
```

Site7_Router — Site 7 subnet 172.31.0.0/25

```
hostname Site7_Router
interface GigabitEthernet0/0
ip address 172.31.0.1 255.255.255.128
duplex auto
speed auto
no shutdown
interface GigabitEthernet0/1
ip address 172.31.1.164 255.255.255.248
duplex auto
speed auto
no shutdown
ip classless
ip route 0.0.0.0 0.0.0.0 172.31.1.161
end
write memory
```

4.5 Switch Configuration – Full (All 10 Switches)

All 10 switches are configured with a VLAN 1 management IP from their respective subnet and a default gateway pointing to the connected router interface. **VLAN 1 is the factory-default VLAN on every Cisco 2960 switch. No additional VLANs, trunk ports, or VTP were configured.**

Aggregation Switches and Data Centre Switch

! ===== Switch1 — Transit SW1 subnet 172.31.1.152/29 =====

```
hostname Switch1
interface Vlan1
ip address 172.31.1.158 255.255.255.248
no shutdown
ip default-gateway 172.31.1.153
end
write memory
```

! ===== Switch0 — Transit SW0 subnet 172.31.1.160/29 =====

```
hostname Switch0
interface Vlan1
ip address 172.31.1.165 255.255.255.248
no shutdown
ip default-gateway 172.31.1.161
end
write memory
```

! ===== Switch2 — Data Centre subnet 172.31.1.128/29 =====

```
hostname Switch2
interface Vlan1
ip address 172.31.1.130 255.255.255.248
no shutdown
ip default-gateway 172.31.1.129
end
write memory
```

Site Access Switches (one per site)

! ===== Site1_Switch — Site 1 subnet 172.31.1.64/27 =====

```
hostname Site1_Switch
interface Vlan1
ip address 172.31.1.66 255.255.255.224
no shutdown
ip default-gateway 172.31.1.65
end
write memory
```

! ===== Site2_Switch — Site 2 subnet 172.31.1.0/27 =====

```
hostname Site2_Switch
interface Vlan1
```

```
ip address 172.31.1.2 255.255.255.224
```

```
no shutdown
```

```
ip default-gateway 172.31.1.1
```

```
end
```

```
write memory
```

```
! ===== Site3_Switch — Site 3 subnet 172.31.0.192/26 =====
```

```
hostname Site3_Switch
```

```
interface Vlan1
```

```
ip address 172.31.0.194 255.255.255.192
```

```
no shutdown
```

```
ip default-gateway 172.31.0.193
```

```
end
```

```
write memory
```

```
! ===== Site4_Switch — Site 4 subnet 172.31.1.96/27 =====
```

```
hostname Site4_Switch
```

```
interface Vlan1
```

```
ip address 172.31.1.98 255.255.255.224
```

```
no shutdown
```

```
ip default-gateway 172.31.1.97
```

```
end
```

```
write memory
```

! ===== Site5_Switch — Site 5 subnet 172.31.0.128/26 =====

```
hostname Site5_Switch
interface Vlan1
ip address 172.31.0.130 255.255.255.192
no shutdown
ip default-gateway 172.31.0.129
end
write memory
```

! ===== Site6_Switch — Site 6 subnet 172.31.1.32/27 =====

```
hostname Site6_Switch
interface Vlan1
ip address 172.31.1.34 255.255.255.224
no shutdown
ip default-gateway 172.31.1.33
end
write memory
```

! ===== Site7_Switch — Site 7 subnet 172.31.0.0/25 =====

```
hostname Site7_Switch
interface Vlan1
ip address 172.31.0.2 255.255.255.128
no shutdown
ip default-gateway 172.31.0.1
end
write memory
```

5. Connectivity Testing and Results

5.1 Testing Matrix

All tests were conducted from the PC's Desktop → Command Prompt in Packet Tracer.

Tests marked ★ are explicitly required by specification. All tests achieved 0% packet loss.

#	Test Type	Source Device	Source IP	Destination	Destination IP	Expected	Actual
1	Intra-Site	Site1_PC1	172.31.1.68	Site1_PC2	172.31.1.69	Success	✓ Pass
2	Intra-Site	Site5_PC1	172.31.0.132	Site5_PC2	172.31.0.133	Success	✓ Pass
3	Inter-Site ★	Site1_PC1	172.31.1.68	Site5_PC1	172.31.0.132	Success	✓ Pass
4	Inter-Site	Site3_PC1	172.31.0.195	Site7_PC1	172.31.0.3	Success	✓ Pass
5	Inter-Site	Site4_PC1	172.31.1.99	Site6_PC1	172.31.1.35	Success	✓ Pass
6	PC → Server ★	Site1_PC1	172.31.1.68	Server1	172.31.1.131	Success	✓ Pass
7	PC → Server ★	Site5_PC1	172.31.0.132	Server1	172.31.1.131	Success	✓ Pass
8	PC → Server	Site7_PC1	172.31.0.3	Server2	172.31.1.132	Success	✓ Pass

5.2 Test Screenshots

Screenshots for all 8 connectivity tests are presented below. Each screenshot shows the PC hostname (visible in the window title bar), the ping command, and the complete statistics line confirming 0% packet loss.

Test 1: Intra-site — Site1_PC1 (172.31.1.68) → Site1_PC2 (172.31.1.69)

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:F7FF:FE16:8DC1
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.31.1.68
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                   172.31.1.65

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 172.31.1.69

Pinging 172.31.1.69 with 32 bytes of data:

Reply from 172.31.1.69: bytes=32 time<1ms TTL=128
Reply from 172.31.1.69: bytes=32 time<1ms TTL=128
Reply from 172.31.1.69: bytes=32 time<1ms TTL=128
Reply from 172.31.1.69: bytes=32 time<1ms TTL=128

Ping statistics for 172.31.1.69:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Test 2: Intra-site — Site5_PC1 (172.31.0.132) → Site5_PC2 (172.31.0.133)

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20D:BDFF:FE83:9142
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.31.0.132
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   172.31.0.129

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 172.31.0.133

Pinging 172.31.0.133 with 32 bytes of data:

Reply from 172.31.0.133: bytes=32 time<1ms TTL=128
Reply from 172.31.0.133: bytes=32 time<1ms TTL=128
Reply from 172.31.0.133: bytes=32 time<1ms TTL=128
Reply from 172.31.0.133: bytes=32 time<1ms TTL=128

Ping statistics for 172.31.0.133:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```


Test 3: Inter-site ★ — Site1_PC1 (172.31.1.68) → Site5_PC1 (172.31.0.132)

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:F7FF:FE16:8DC1
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.31.1.68
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                172.31.1.65

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 172.31.0.132

Pinging 172.31.0.132 with 32 bytes of data:

Reply from 172.31.0.132: bytes=32 time=1ms TTL=124
Reply from 172.31.0.132: bytes=32 time=1ms TTL=124
Reply from 172.31.0.132: bytes=32 time=1ms TTL=124
Reply from 172.31.0.132: bytes=32 time=1ms TTL=124

Ping statistics for 172.31.0.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>|
```

Test 4: Inter-site — Site3_PC1 (172.31.0.195) → Site7_PC1 (172.31.0.3)

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:BCFF:FE13:C330
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 172.31.0.195
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   172.31.0.193

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 172.31.0.3

Pinging 172.31.0.3 with 32 bytes of data:

Reply from 172.31.0.3: bytes=32 time=3ms TTL=124
Reply from 172.31.0.3: bytes=32 time=20ms TTL=124
Reply from 172.31.0.3: bytes=32 time=20ms TTL=124
Reply from 172.31.0.3: bytes=32 time=12ms TTL=124

Ping statistics for 172.31.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 20ms, Average = 13ms

C:\>
```

Test 5: Inter-site — Site4_PC1 (172.31.1.99) → Site6_PC1 (172.31.1.35)

```
C:\>
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20B:BEFF:FE0A:13D1
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.31.1.99
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                   172.31.1.97

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 172.31.1.35

Pinging 172.31.1.35 with 32 bytes of data:

Reply from 172.31.1.35: bytes=32 time=1ms TTL=124
Reply from 172.31.1.35: bytes=32 time=1ms TTL=124
Reply from 172.31.1.35: bytes=32 time=10ms TTL=124
Reply from 172.31.1.35: bytes=32 time=1ms TTL=124

Ping statistics for 172.31.1.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 3ms

C:\>
```

Test 6: PC→Server ★— Site1_PC1 (172.31.1.68) → Server1 (172.31.1.131)

```
C:\>
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:F7FF:FE16:8DC1
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 172.31.1.68
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                172.31.1.65

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 172.31.1.131

Pinging 172.31.1.131 with 32 bytes of data:

Reply from 172.31.1.131: bytes=32 time=1ms TTL=125
Reply from 172.31.1.131: bytes=32 time=1ms TTL=125
Reply from 172.31.1.131: bytes=32 time=1ms TTL=125
Reply from 172.31.1.131: bytes=32 time=7ms TTL=125

Ping statistics for 172.31.1.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 2ms

C:\>
```

Test 7: PC→Server ★— Site5_PC1 (172.31.0.132) → Server1 (172.31.1.131)

```
C:\>
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20D:BDFF:FE83:9142
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.31.0.132
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   172.31.0.129

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 172.31.1.131

Pinging 172.31.1.131 with 32 bytes of data:

Reply from 172.31.1.131: bytes=32 time<1ms TTL=126
Reply from 172.31.1.131: bytes=32 time=1ms TTL=126
Reply from 172.31.1.131: bytes=32 time<1ms TTL=126
Reply from 172.31.1.131: bytes=32 time<1ms TTL=126

Ping statistics for 172.31.1.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Test 8: PC→Server — Site7_PC1 (172.31.0.3) → Server2 (172.31.1.132)

```
C:\>
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::202:16FF:FE1D:CB6
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.31.0.3
    Subnet Mask . . . . .: 255.255.255.128
    Default Gateway . . . . .: ::
                                   172.31.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 172.31.1.132

Pinging 172.31.1.132 with 32 bytes of data:

Reply from 172.31.1.132: bytes=32 time<1ms TTL=126
Reply from 172.31.1.132: bytes=32 time<1ms TTL=126
Reply from 172.31.1.132: bytes=32 time<1ms TTL=126
Reply from 172.31.1.132: bytes=32 time<1ms TTL=126

Ping statistics for 172.31.1.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

6. Design Reflection

6.1 Why VLSM was used and how Subnet Sizes were determined

VLSM was selected over classful equal-size subnetting because of the significant variation in user counts across the seven sites, ranging from 14 users at Site 4 to 80 users at Site 7 - a difference of more than five-fold. Applying a uniform /25 subnet to every site (required to accommodate Site 7) would allocate 126 usable addresses per location regardless of actual need. Sites 4 and 1 would each receive 126 addresses but use only 16 and 23 respectively - a waste rate of up to 87% at Site 4 and over 80% at Site 1. Across all seven sites this would consume $7 \times 128 = 896$ addresses for LAN subnets alone, and with transit subnets, Data Centre, and P2P links added, the demand would exceed 900 addresses. While this technically fits within the 8,190 usable addresses of the /19 block, it represents severe inefficiency and leaves minimal headroom for future growth. By contrast, the VLSM plan allocates subnets largest-first starting at 172.31.0.0, ensuring no gaps or overlaps across all 13 subnets within the assigned block.

VLSM resolves this by allocating the smallest subnet that satisfies each site's specific host requirement. Subnet sizes were determined by calculating the total mandatory address demand per site: user devices plus one gateway IP (site router LAN interface) plus one switch management IP (required on VLAN 1 per Section 6c of the specification). This calculation directly determined prefix length - a /25 for Site 7 (82 hosts needed), /26 for Sites 5 and 3 (31 and 30 hosts needed), and /27 for all remaining sites (16-26 hosts needed). The result is a tightly efficient plan consuming only 420 addresses across all 13 subnets, with more than 7,700 addresses remaining in reserve within the assigned block.

6.2 MAC Addressing vs IP Addressing – Packet Delivery from Site 1 to the Data Centre

When Site1_PC1 (172.31.1.68) sends a packet to Server1 (172.31.1.131) in the Data Centre, two complementary addressing systems operate simultaneously at different layers of the OSI model, each serving a distinct purpose in delivering the packet.

IP addressing operates at Layer 3 and provides logical, end-to-end addressing. The source IP (172.31.1.68) and destination IP (172.31.1.131) are embedded in the IP header and remain absolutely unchanged for the entire journey from Site1_PC1 to Server1, regardless of how many router hops the packet traverses. At each router, the IP destination address is compared against the routing table to determine the correct outgoing interface and next hop. Site1_Router has a default route pointing to Router1 (172.31.1.153), so the packet is forwarded to Router1 via the Transit SW1 segment. On Router1, the destination 172.31.1.131 matches the static route for 172.31.1.128/29 pointing to Router2 via the serial link (172.31.1.146). On Router2, 172.31.1.131 falls within the directly connected Data Centre subnet on Gi0/1, so the packet is forwarded to Switch2 and delivered to Server1. This three-hop path is confirmed by the TTL=125 observed in connectivity testing (Windows default TTL 128 minus 3 decrements).

MAC addressing operates at Layer 2 and provides physical, hop-by-hop addressing within each network segment. Unlike IP addresses, the MAC addresses in the frame header are replaced at every router boundary. Site1_PC1 first sends an ARP request to discover Site1_Router's MAC address (the default gateway 172.31.1.65). It then encapsulates the packet in a frame with its own MAC as source and Site1_Router's MAC as destination. When Site1_Router receives this frame, it strips the Layer 2 header, examines the Layer 3 IP destination, and creates a new Ethernet frame with its transit interface MAC as source and Router1's MAC as destination on the Switch1 segment. Router1 repeats this process, forwarding the packet toward Router2 on the serial link. Router2 then creates a final frame addressed to Server1's MAC on the Data Centre LAN. The IP addresses stayed constant across all three hops; the MAC addresses changed at every router boundary - a clear illustration of the distinct roles of Layer 2 and Layer 3 addressing in a routed multi-site network.

6.3 Switched LAN vs Hub-Based Design – Efficiency for Multi-Site Offices

A network hub operates at OSI Layer 1 (Physical) and functions as a simple electrical signal repeater: every frame received on any port is immediately retransmitted to all other ports. This creates a single collision domain across the entire hub segment. In the context of a branch office with 21 or more user devices, this means every transmission collides with every other simultaneous transmission. When two devices transmit simultaneously, both frames are corrupted, both senders must wait a random backoff period, and then retransmit. As the number of users increases, collision frequency rises non-linearly, causing exponential growth in retransmissions and a corresponding collapse in effective throughput. Hubs operate in half-duplex mode, meaning no port can transmit and receive simultaneously, which halves the theoretical maximum data rate available to each connected device. For an Education Services environment where staff and students may simultaneously access learning management systems, stream educational content, or transfer large assignment files, a hub-based design would produce unacceptably poor performance.

A Cisco 2960-24TT switch operates at OSI Layer 2 (Data Link) and maintains a MAC address table built from inspecting the source MAC address of every incoming frame. When a frame arrives destined for a known MAC address, the switch forwards it only to the specific port associated with that MAC - no other port receives the frame. Each switch port therefore forms its own independent collision domain, eliminating the inter-device collisions that plague hub networks. Furthermore, modern switches support full-duplex operation on every port, allowing simultaneous transmission and reception and doubling the effective bandwidth available per connection. In this network, Site 7 alone has 80 users, each receiving a dedicated 100 Mbps full-duplex connection through the Cisco 2960-24TT. The switch's forwarding decision - made in hardware using the MAC table - occurs at wire speed with negligible delay, delivering consistent performance regardless of how many other users are active on adjacent ports.

6.4 Triangle Core vs Linear Chain – One Advantage and One Challenge

Advantage → Path redundancy and fault tolerance: The triangle topology connecting MainRouter, Router1, and Router2 via three independent links means that a failure of any single link does not isolate any part of the network. If, for example, the GigabitEthernet link between MainRouter and Router2 (172.31.1.140/30) fails, traffic from Sites 5-7 and the Data Centre can still reach Sites 1-4 by traversing the Router2 → Router1 serial link and then the Router1 → MainRouter link. No single point of link failure can partition the network. By contrast, a linear chain topology (Main Router Router1 → Router2) would completely isolate Router2 and all devices behind it - Sites 5-7 and the Data Centre - if the Router1-Router2 link failed. For an Education Services organization where server access is business-critical, this redundancy provides meaningful protection against link failures during operational hours.

Challenge → Static routing complexity and loop prevention: The triangle topology introduces routing design complexity that is absent in a linear chain. With three routers each connected to two peers, the routing table on each router must cover all 13 subnets via specific next-hops, and the designer must carefully ensure that no circular routes exist. In this project, MainRouter requires 11 static routes, Router1 requires 10, and Router2 requires 9 -substantially more than the 4-5 needed in a linear chain. More critically, incorrect route configuration in a triangle can create routing loops: for example, if Router1 sends packets for an unknown subnet to MainRouter and MainRouter returns them to Router1, the packets will loop until their TTL expires, consuming bandwidth and preventing delivery. Preventing this required thorough verification using show IP route on all three routers to confirm that every subnet appears exactly once in each routing table with a sensible next-hop, and that no two routers are pointing at each other for the same destination.

Appendix – Full Device Addressing Table

Complete IP addressing information for every device and interface in the network. All addresses fall within the assigned 172.31.0.0/19 block with no overlaps.

Device	Interface	IP Address	Pfx	Subnet Mask	Gateway	Notes
MainRouter	GigabitEthernet0/0	172.31.1.137	/30	255.255.255.252	172.31.1.138	R0↔R1 link
MainRouter	GigabitEthernet0/1	172.31.1.141	/30	255.255.255.252	172.31.1.142	R0↔R2 link
Router1	GigabitEthernet0/0/0	172.31.1.138	/30	255.255.255.252	172.31.1.137	R1↔R0 link
Router1	GigabitEthernet0/0/1	172.31.1.153	/29	255.255.255.248	—	Transit SW1 gateway
Router1	Serial0/1/0	172.31.1.145	/30	255.255.255.252	172.31.1.146	R1↔R2 serial
Router2	GigabitEthernet0/0	172.31.1.142	/30	255.255.255.252	172.31.1.141	R2↔R0 link
Router2	GigabitEthernet0/1	172.31.1.129	/29	255.255.255.248	—	Data Centre gateway
Router2	GigabitEthernet0/2	172.31.1.161	/29	255.255.255.248	—	Transit SW0 gateway
Router2	Serial0/0/0	172.31.1.146	/30	255.255.255.252	172.31.1.145	R2↔R1 serial
Site1_Router	GigabitEthernet0/0	172.31.1.657	/27	255.255.255.224	172.31.1.65	Site 1 LAN gateway
Site1_Router	GigabitEthernet0/1	172.31.1.154	/29	255.255.255.248	172.31.1.153	Transit SW1
Site2_Router	GigabitEthernet0/0	172.31.1.17	/27	255.255.255.224	172.31.1.1	Site 2 LAN gateway
Site2_Router	GigabitEthernet0/1	172.31.1.155	/29	255.255.255.248	172.31.1.153	Transit SW1
Site3_Router	GigabitEthernet0/0	172.31.0.193	/26	255.255.255.192	172.31.0.193	Site 3 LAN gateway
Site3_Router	GigabitEthernet0/1	172.31.1.156	/29	255.255.255.248	172.31.1.153	Transit SW1
Site4_Router	GigabitEthernet0/0	172.31.1.97	/27	255.255.255.224	172.31.1.97	Site 4 LAN gateway
Site4_Router	GigabitEthernet0/1	172.31.1.157	/29	255.255.255.248	172.31.1.153	Transit SW1
Site5_Router	GigabitEthernet0/0	172.31.0.129	/26	255.255.255.192	172.31.0.129	Site 5 LAN gateway

Site5_Router	GigabitEthernet0/1	172.31.1.162	/29	255.255.255.248	172.31.1.161	Transit SW0
Site6_Router	GigabitEthernet0/0	172.31.1.33	/27	255.255.255.224	172.31.1.33	Site 6 LAN gateway
Site6_Router	GigabitEthernet0/1	172.31.1.163	/29	255.255.255.248	172.31.1.161	Transit SW0
Site7_Router	GigabitEthernet0/0	172.31.0.1	/25	255.255.255.128	172.31.0.1	Site 7 LAN gateway
Site7_Router	GigabitEthernet0/1	172.31.1.164	/29	255.255.255.248	172.31.1.161	Transit SW0
Switch1	VLAN 1	172.31.1.158	/29	255.255.255.248	172.31.1.153	Management - Transit SW1
Switch0	VLAN 1	172.31.1.165	/29	255.255.255.248	172.31.1.161	Management - Transit SW0
Switch2	VLAN 1	172.31.1.130	/29	255.255.255.248	172.31.1.129	Management - DC subnet
Site1_Switch	VLAN 1	172.31.1.66	/27	255.255.255.224	172.31.1.65	Management - Site 1
Site2_Switch	VLAN 1	172.31.1.2	/27	255.255.255.224	172.31.1.1	Management - Site 2
Site3_Switch	VLAN 1	172.31.0.194	/26	255.255.255.192	172.31.0.193	Management - Site 3
Site4_Switch	VLAN 1	172.31.1.98	/27	255.255.255.224	172.31.1.97	Management - Site 4
Site5_Switch	VLAN 1	172.31.0.130	/26	255.255.255.192	172.31.0.129	Management - Site 5
Site6_Switch	VLAN 1	172.31.1.34	/27	255.255.255.224	172.31.1.33	Management - Site 6
Site7_Switch	VLAN 1	172.31.0.2	/25	255.255.255.128	172.31.0.1	Management - Site 7
Site1_PC1	FastEthernet0	172.31.1.68	/27	255.255.255.224	172.31.1.65	
Site1_PC2	FastEthernet0	172.31.1.69	/27	255.255.255.224	172.31.1.65	
Site1_PC3	FastEthernet0	172.31.1.70	/27	255.255.255.224	172.31.1.65	
Site2_PC1	FastEthernet0	172.31.1.3	/27	255.255.255.224	172.31.1.1	
Site2_PC2	FastEthernet0	172.31.1.4	/27	255.255.255.224	172.31.1.1	
Site2_PC3	FastEthernet0	172.31.1.5	/27	255.255.255.224	172.31.1.1	
Site3_PC1	FastEthernet0	172.31.0.195	/26	255.255.255.192	172.31.0.193	
Site3_PC2	FastEthernet0	172.31.0.196	/26	255.255.255.192	172.31.0.193	
Site3_PC3	FastEthernet0	172.31.0.197	/26	255.255.255.192	172.31.0.193	

Site4_PC1	FastEthernet0	172.31.1.99	/27	255.255.255.224	172.31.1.97	
Site4_PC2	FastEthernet0	172.31.1.100	/27	255.255.255.224	172.31.1.97	
Site4_PC3	FastEthernet0	172.31.1.101	/27	255.255.255.224	172.31.1.97	
Site5_PC1	FastEthernet0	172.31.0.132	/26	255.255.255.192	172.31.0.129	
Site5_PC2	FastEthernet0	172.31.0.133	/26	255.255.255.192	172.31.0.129	
Site5_PC3	FastEthernet0	172.31.0.134	/26	255.255.255.192	172.31.0.129	
Site6_PC1	FastEthernet0	172.31.1.35	/27	255.255.255.224	172.31.1.33	
Site6_PC2	FastEthernet0	172.31.1.36	/27	255.255.255.224	172.31.1.33	
Site6_PC3	FastEthernet0	172.31.1.37	/27	255.255.255.224	172.31.1.33	
Site7_PC1	FastEthernet0	172.31.0.3	/25	255.255.255.128	172.31.0.1	
Site7_PC2	FastEthernet0	172.31.0.4	/25	255.255.255.128	172.31.0.1	
Site7_PC3	FastEthernet0	172.31.0.5	/25	255.255.255.128	172.31.0.1	
Server1	FastEthernet0	172.31.1.131	/29	255.255.255.248	172.31.1.129	
Server2	FastEthernet0	172.31.1.132	/29	255.255.255.248	172.31.1.129	

Reflective on AI Use

Two AI tools were used in completing this project.

Claude (Anthropic, claude.ai) was used during the documentation process — identifying errors in the report and assisting with drafting and refining the writing in the report.

QuillBot was used to assist with paraphrasing and improving sentence structure in selected sections of the report. All final decisions regarding design choices, subnet allocation, and configuration were made by me and verified against the actual Packet Tracer implementation.

All technical implementation work — building the topology in Cisco Packet Tracer, configuring every router and switch via the CLI, running connectivity tests, and taking screenshots — was performed independently.